

Mary Dahl, Ph.D.

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OBJECTIVE

Recent Ph.D. graduate and satellite engineer working on autonomous satellite systems, data processing for remote sensing, and simulation design. Brings a broad set of engineering skills, including test engineering, safety analysis, and software development, with strong ability to rapidly learn new topics and techniques. Looking to expand skills in instrumentation design, software, and testing for scientific missions.

EDUCATION

Massachusetts Institute of Technology, Cambridge, MA

Ph.D. in Aerospace Engineering

Jun. 2022 – Feb. 2026

Thesis: Autonomous and Dynamic Satellite for Improved Cloud Observation

GPA: 4.9 / 5.0

Master of Science in Aerospace Engineering

Jun. 2020 – Jun. 2022

Thesis: Structures and Methods for Safe On Orbit Robotic Assembly of Small Satellites

GPA: 4.9 / 5.0

Bachelor of Science in Aerospace Engineering with a minor in Creative Writing

Aug. 2016 – May. 2020

GPA: 4.9 / 5.0

RESEARCH EXPERIENCE

Dynamic Targeting and Intelligent Instrumentation for Cloud Science, STAR Lab *Oct. 2023 – Dec. 2025*

- Developed mission concept for onboard decision making and autonomous pointing for efficient maximization of scientific data collection.
- Optimized instrument selection, such as lidar, radar, and optical systems, for cloud research by analyzing scientific value, power requirements, cost, and suitability for dynamic measurement with a genetic algorithm.
- Simulated the performance of an off-nadir lidar to quantify performance drop off, identifying 30 degrees as maximum off-nadir angle.
- Built simulation environment to model satellite systems and to evaluate three different cloud prioritization algorithms performances, with the best algorithm measuring over 16% improvement over state-of-the-art.

SPECIES: Smart Polyimide Expandable Collector, STAR Lab

Jun. 2024 – Aug. 2025

- Developed models for assessing the signal quality of a diffractive optical element as a lidar receiver payload on a small satellite for measuring cloud top height.

SEASALT: Satellite for Estimating Aquatic Salinity and Temperature, STAR Lab

May. 2021 – May. 2023

- Managed a multi-institutional team (MIT, WHOI, NRL) to design a CubeSat to measure ocean salinity with a custom optical system, onboard calibrators, and regression algorithms.
- Wrote requirements, technical budgets, and interface control documents for MDR and sPDR reviews.
- Performed all structural design and simulations, including thermal and vibrational analysis.

Robotic CubeSat Assembly, STAR Lab

Jan. 2020 – May. 2022

- Designed and tested 3D-printed, compliantly designed, snap fit shells for CubeSat components that could be manipulated and assembled by a cartesian robot safely and efficiently in less than 4 minutes.
- Wrote and implemented a comprehensive safety analysis using System-Theoretic Process Analysis (STPA).
- Managed a team of undergraduate and graduate students to design test electronics, program the robot, perform extensive hardware testing, and conduct anomaly investigation.

WORK EXPERIENCE

MIT Communication Lab, Cambridge, MA

Feb. 2024 – Present.

Communication Fellow

- Developed technical communication resources for undergraduate and graduate students, including how-to kits (2), blog posts (2), posters (1), and workshops (9; 100+ participants).
- Coached 60+ one-on-one appointments with students on improving technical communication, including presentations, fellowship applications, resumes, and papers.

MIT Lincoln Laboratory Beaverworks Summer Institute, Lexington, MA

2020, 2023

TA, Lead Instructor

- Managed a class of 19 high school students for a full day summer program to teach them the basics of satellite engineering, Python, and the use of amateur hardware such as Raspberry Pis, IMUs, radios, and solar panels.
- Developed and taught lectures, wrote and implemented interactive labs to reinforce learning, and mentored students through an open ended final project to build a functional miniature satellite.

SpaceX, Hawthorne, CA

Jun. 2019 – Aug. 2019

Summer Intern

- Developed and executed a large-scale pyrotechnics test as part of the Development Test Engineering Team to finalize the inclusion of a critical safety feature in the Dragon Crew Capsule.
- Organized assembly and instrumentation of a custom test stand, planned and oversaw a large-scale lift operation, and coordinated with three teams of stakeholders, successfully qualifying performance.

MIT Lincoln Laboratory, Lexington, MA

Jun. 2018 – Aug. 2018

Summer Intern

- Designed and fabricated a test stand for verifying performance of experimental microbolometer IR cameras in different weather conditions and temperatures for the Tactical Defense Systems Group.
- Verified the performance of cameras by performing baseline hardware testing and blackbody characterization.

LEADERSHIP

Queer Advocacy Space in AeroAstro, Cambridge, MA

Mar. 2023 – Present.

Executive Board Member

- Advocated for the needs of queer students in the department by hosting teaching events to improve department-wide awareness and social events for community building.

Graduate Association of Aeronautics and Astronautics, Cambridge, MA

Jan. 2021 – Jan. 2024

Social Chair, Graduate Student Council Representative

- Planned and ran social events for approximately 300 graduate students, including off-campus trips, office trick-or-treating, coffee socials, and new student mixers.
- Worked with the interdisciplinary Graduate Student Council to advocate for the needs of AeroAstro students during the transition back to in-person instruction and activities.

MIT Shakespeare Ensemble, Cambridge, MA

Jan. 2017 – May. 2020

Technical Director, Producer, Actor

- Recruited and lead a team of 20+ students to put together productions of theater shows by managing requirements, deadlines, and conflicts.
- Coordinated the full-day technical setup by managing students across multiple teams to complete set construction, lighting and sound setup, audience arrangements, and backstage preparations.

MIT Next Haunt, Cambridge, MA

Sept. 2016 – Nov. 2019

Producer, Assistant Producer, Assistant Build Director

- Lead a team to design and construct a wooden, two-story haunted house escape room, featuring crawl tunnels, props for implementing jump scares, and electronic puzzles.
- Conducted operation of event for multiple weekends by coordinating “zombie” actors, properly timing hint delivery for optimal experiences, and rapidly responding to unexpected problems.

HONORS

MathWorks Fellowship (awarded twice)

2023, 2024

MIT AeroAstro Outstanding Graduate Student Leadership Recognition Award

May 2023

Actor for The V.E.R.S. Project, 2022 Webby Award in Best Social Gaming Experience

May 2022

MIT AeroAstro Community Recognition Graduate Teaching Assistantship Award

April 2022

PROFESSIONAL DEVELOPMENT

NASA Summer School: Using Satellite Observations to Advance Climate Models

Aug. 2023

Gordon-MIT Engineering Leadership Program

Sept. 2018 - Mar. 2019

SKILLS

Autonomy & Algorithms: Task prioritization, machine learning for image processing, optimization, genetic algorithms, simulation-based validation

Modeling & Simulation: Satellite system and sensor performance modeling, STK, MATLAB, Python

Satellite Data Processing & Geospatial Analysis: Remote sensing data pipelines, NetCDF, xarray, large-scale dataset handling, statistical analysis and visualization

Software & Computing: Python (scientific computing), MATLAB, Linux, Git-based workflows, L^AT_EX

Instrumentation & Test: Optical payloads, test planning, safety analysis (STPA), requirements verification

Mechanical Design & Fabrication: SolidWorks, Ansys Mechanical, tolerance-driven design, rapid prototyping, laser cutting, 3D printing, machine shop fabrication